

Sensation and Perception

Psychology Notes

1. Concept and Definition of Sensation

Sensation is the process through which sensory organs receive stimulation from the external or internal environment and convert it into neural signals that can be transmitted to the brain.

Definition

Sensation may be defined as the process by which stimulation of sensory receptors produces neural impulses that represent experiences occurring inside or outside the body.

Sensation is considered the first step in acquiring information about the environment. Without sensation, perception cannot occur.

Characteristics of Sensation

- It is a basic psychological process.
- It depends upon stimulation of sensory receptors.
- It involves the conversion of physical energy into neural energy.
- It provides raw information to the brain.
- It is influenced by the intensity and duration of stimuli.

2. Types of Sensation

Psychologists generally classify sensations into three broad categories.

A. Exteroceptive Sensations

These sensations provide information about the external environment.

Visual Sensation

- Stimulus: Light waves
- Receptor: Retina of the eye
- Function: Enables vision, color recognition, and shape identification.

Auditory Sensation

- Stimulus: Sound waves
- Receptor: Hair cells in the cochlea
- Function: Hearing and speech perception.

Olfactory Sensation

- Stimulus: Airborne chemical substances
- Receptor: Olfactory receptors in the nose
- Function: Smell perception.

Gustatory Sensation

- Stimulus: Chemicals dissolved in saliva
- Receptor: Taste buds

- Function: Taste perception.

Basic tastes include:

- Sweet
- Sour
- Salty
- Bitter
- Umami

Cutaneous Sensation

Includes sensations arising from the skin.

Types:

- Touch
- Pressure
- Warmth
- Cold
- Pain

B. Proprioceptive Sensations

These sensations provide information about body position and movement.

Kinesthetic Sensation

- Receptors are present in muscles, tendons, and joints.
- Helps coordinate voluntary movements.

Example: Walking without looking at one's feet.

Vestibular Sensation

- Receptors are located in the inner ear.
- Responsible for balance and spatial orientation.

C. Interoceptive Sensations

These sensations arise from internal organs.

Examples:

- Hunger
- Thirst
- Nausea
- Fullness
- Internal pain

These sensations help maintain homeostasis.

3. Sensory Receptors

Sensory receptors are specialized cells that detect environmental changes and transform physical energy into neural impulses.

Types of Sensory Receptors

1. Photoreceptors

- Respond to light.
- Location: Retina of the eye.
- Examples: Rods, Cones
- Functions: Vision and color perception.

2. Mechanoreceptors

- Respond to mechanical stimulation.
- Examples: Touch receptors, Hearing receptors, Pressure receptors
- Functions: Touch, vibration, hearing, balance.

3. Chemoreceptors

- Respond to chemical substances.
- Examples: Taste buds, Olfactory receptors
- Functions: Taste and smell.

4. Thermoreceptors

- Respond to temperature changes.
- Functions: Detect heat and cold.

5. Nociceptors

- Respond to harmful stimuli.
- Function: Pain perception.

4. Sensory Thresholds

A sensory threshold refers to the minimum amount of stimulation required for detection.

A. Absolute Threshold

The smallest amount of stimulus that can be detected 50% of the time.

Examples:

- Seeing a candle flame from a long distance.
- Hearing a faint whisper.

Factors affecting absolute threshold:

- Fatigue
- Motivation
- Practice
- Expectations

B. Difference Threshold

Also called **Just Noticeable Difference (JND)**.

It is the minimum difference between two stimuli that can be detected 50% of the time.

Example: Noticing a slight increase in the brightness of a bulb.

Weber's Law

It states that the just noticeable difference is a constant proportion of the original stimulus intensity.

Formula: $\Delta I / I = K$

Where:

- ΔI = Increment threshold
- I = Initial stimulus intensity
- K = Constant

5. Sensory Adaptation

Sensory adaptation is the decrease in sensitivity to a constant and unchanging stimulus over time. It allows individuals to focus on new and important environmental changes.

Examples

- No longer noticing the ticking of a clock.
- Becoming unaware of the smell of perfume after a few minutes.
- Adjusting to cold water after staying in it for some time.

Importance

- Prevents sensory overload.
- Conserves attentional resources.
- Increases responsiveness to novel stimuli.

6. Meaning of Attention

Attention is the process of selectively concentrating consciousness on certain stimuli while ignoring others.

Definition

Attention may be defined as the focusing of mental activity on a particular object, event, or thought.

Characteristics

- Selective in nature.
- Limited in capacity.
- Dynamic and shifting.
- Essential for learning and memory.

Factors Affecting Attention

Objective Factors

- Intensity
- Size
- Contrast
- Movement
- Novelty

Subjective Factors

- Interest
- Motivation
- Emotional state
- Habits

7. Meaning of Perception

Perception is the process through which the brain organizes, interprets, and gives meaning to sensory information.

Definition

Perception may be defined as the process of selecting, organizing, and interpreting sensory input to understand the environment.

Perception differs from sensation because sensation provides raw data, whereas perception gives meaning to that data.

Example

Sensation: Light enters the eyes.

Perception: Recognizing the object as a book.

8. Perceptual Organization

Perceptual organization refers to the tendency of the brain to arrange sensory stimuli into meaningful patterns.

Psychologists associated with the Gestalt approach proposed several principles.

Principles of Perceptual Organization

1. Figure-Ground Relationship

Individuals distinguish an object (figure) from its background (ground).

Example: Reading black letters printed on white paper.

2. Proximity

Objects located near one another are perceived as belonging together.

Example: Students sitting close together are perceived as a group.

3. Similarity

Similar objects tend to be grouped together.

Example: People wearing the same uniform appear to belong to one team.

4. Closure

People tend to perceive incomplete figures as complete.

Example: Recognizing a broken circle as a full circle.

5. Continuity

Stimuli arranged in a continuous pattern are perceived as related.

Example: Perceiving crossing lines as smooth uninterrupted lines.

6. Common Fate

Objects moving in the same direction are perceived as a group.

Example: Birds flying together in a flock.

9. Perceptual Constancies

Perceptual constancy refers to the tendency to perceive objects as stable despite changes in sensory input.

A. Size Constancy

Objects are perceived as maintaining the same size despite changes in distance.

Example: A person walking away appears smaller on the retina but is perceived as having the same actual size.

B. Shape Constancy

Objects retain their perceived shape even when viewed from different angles.

Example: A door appears rectangular whether open or closed.

C. Brightness Constancy

Objects are perceived as having consistent brightness under different lighting conditions.

Example: White paper appears white in sunlight and indoors.

D. Color Constancy

Objects are perceived as having stable colors despite changes in illumination.

Example: A red apple appears red both in daylight and under indoor lighting.

Summary

Sensation is the process of detecting environmental stimuli through sensory receptors, while perception is the process of organizing and interpreting these sensory experiences. Sensory thresholds determine the limits of detection, and sensory adaptation helps individuals adjust to constant stimulation. Attention enables selective focus, whereas perceptual organization and perceptual constancies allow people to perceive the world as coherent, stable, and meaningful. This integrated functioning of sensation, attention, and perception forms the basis of human interaction with the environment.